

Federico Bello

Machine Learning Engineer | MVA @ ENS Paris-Saclay, Sophie Germain Scholar
Paris, France +33 6 88 80 61 55 fe.debello13@gmail.com fedebello.com LinkedIn GitHub

Experience

Lead Machine Learning Engineer — Tryolabs

Oct 2024 – Sep 2026

- Embedded as a Forward Deployed Engineer with the LlamaIndex team, taking document-processing systems from prototype to production for enterprise clients (BP, EY, Guidehouse).
- Technical Lead for Geosphere (UNICEF), an open-source agent system built on the Model Context Protocol, named by IRCAI (UNESCO) among the Top 100 AI projects globally.
- Define engineering standards for AI system design and deployment, establishing robust CI/CD patterns and translating complex methodologies into production-ready software.
- Mentor the engineering team, lead technical hiring and onboarding, and represent the company through public talks and internal workshops.

Teaching and Research Assistant — School of Engineering, UdelaR

Feb 2023 – Feb 2025

- Delivered lectures and labs for Programming I and II (1000+ students), focusing on algorithm design, data structures, and computational complexity.
- Conducted research within the Information Theory Group on DNA storage channels, contributing to theoretical derivations and proofs, and co-authoring peer-reviewed publications.

AI Developer — Eagerworks

Feb 2024 – Oct 2024

- Led backend development of *Docshunter*, an AI-powered document management and generation platform built on LLM pipelines (LangChain).

Junior Developer — Sinapsis

Aug 2023 – Feb 2024

- Developed risk analysis web applications for financial institutions using .NET technologies.

Education

M.Sc. Mathematics, Vision, Learning (MVA)

Sep 2026 – Sep 2027

École Normale Supérieure Paris-Saclay, Paris, France

- Graduate program in applied mathematics for machine learning, computer vision and statistical learning.
- Awarded the Sophie Germain Excellence Scholarship by the Fondation Mathématique Jacques Hadamard.

B.Sc. in Telecommunication Systems Engineering

Mar 2019 – Jun 2025

University of the Republic (UdelaR), Montevideo, Uruguay

- Graduated top 3 of the class; specialization in Machine Learning and Statistical Signal Processing.
- Thesis: *Time Series Anomaly Detection with Graph Neural Networks*, a GNN-based framework for multivariate anomaly detection, evaluating graph construction and benchmark biases. Awarded Best Engineering Project and First Place Thesis Prize by the National Academy of Engineering (Uruguay).
- Author and maintainer of GraGOD, an open-source framework optimizing GNN architectures for high-dimensional time series.

Publications

GNNs for Time Series Anomaly Detection: An Open-Source Framework and a Critical Evaluation

Mar 2026

Int. Conference on Pattern Recognition Applications and Methods (ICPRAM 2026), Marbella, Spain

The Lattice-Input Discrete-Time Poisson Channel

Aug 2024

IEEE International Symposium on Information Theory (ISIT), Athens, Greece

Projects

Rupta — Co-founder & Technical Lead

Mar 2026 – Present

- Cross-platform social app (React Native + web) that gamifies real-world experiences: users complete missions, earn points, and compete across feeds and rankings. Own full-stack architecture; launching on iOS and Android. [rupta.app](#)

Technical Skills

Programming: Python, C++, C#, SQL

Machine Learning: Graph Neural Networks (GNNs), Agentic Workflows, LlamaIndex, LangChain, PyTorch, TensorFlow, Scikit-learn, Langfuse

Cloud & Infrastructure: GCP (Professional ML Engineer, certified), AWS, Docker

Mathematics: Probability, Linear Algebra, Optimization, Information Theory

Languages: Spanish (Native), English (C2 Proficiency, CPE), French (A2), German (B1, Goethe)